

```

1  public class NodeTest {
2
3  public static void main (String[] args)
4  {
5      int [] arr = { 4, 7, 3, 5 };
6      Node<Integer> lst1 = build(arr);
7      Node<Integer> lst2 = buildToNum(8);
8      show (lst1);
9      show(lst2);
10     System.out.println(isSorted(lst2));
11     System.out.println(isSortedReq(lst1));
12 }
13 //--- פוסט הלקחה תרשרש תיינב ---
14 public static Node<Integer> build (int [] arr)
15 {
16     Node<Integer> lst = new
17         Node<Integer>(arr[0]);
18     Node<Integer> pos = lst;
19
20     for (int i = 1 ; i < arr.length ; i++)
21     {
22         pos.setNext (new Node<Integer> (arr[i]));
23         pos = pos.getNext();
24     }
25     return lst;
26 }
27
28 public static void show (Node<Integer> lst)
29 {
30     System.out.print("[");
31     while (lst != null)
32     {
33         System.out.print(lst.toString());
34         if (lst.hasNext())
35             System.out.print(", ");
36         lst = lst.getNext();
37     }
38     System.out.println("]");
39 }
40
41 public static int sumList(Node<Integer> lst){
42     int sum = 0;
43     while (lst != null)
44     {
45         sum += lst.getValue();
46         lst = lst.getNext();
47     }
48
49     return sum;
50 }
51
52 public static Node<Integer> buildToNum (int num)
53 {

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```
54     Node<Integer> lst = new Node<Integer>(0);
55     Node<Integer> pos = lst;
56
57     for (int i = 1 ; i < num ; i++)
58     {
59         pos.setNext (new Node<Integer> (i));
60         pos = pos.getNext();
61     }
62     return lst;
63 }
64
65 public static boolean isNumberInList(int num, Node<Integer> lst){
66     while (lst != null)
67     {
68         if (lst.getValue() == num)
69             return true;
70         lst = lst.getNext();
71     }
72
73     return false;
74 }
75
76 public static boolean isNumberInListReq(int num, Node<Integer> lst){
77     if (lst == null) return false;
78     if (lst.getValue() == num) return true;
79
80     return isNumberInListReq(num, lst.getNext());
81 }
82
83 public static boolean isSorted(Node<Integer> lst){
84     while (lst != null)
85     {
86         Node<Integer> next = lst.getNext();
87         if (next != null){
88             if (lst.getValue() > next.getValue())
89                 return false;
90         }
91         lst = next;
92     }
93     return true;
94 }
95
96 public static boolean isSortedReq(Node<Integer> lst){
97     Node<Integer> next = lst.getNext();
98     if (next == null) return true;
99
100    return (lst.getValue() < next.getValue() && isSortedReq(next));
101 }
102 }
103
```